Kaiwen Hong

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Education

- **University of Illinois at Urbana-Champaign** — Sept. 2021 - Aug. 2026 (Expected)
  Ph.D. Candidate in Electrical and Computer Engineering — *GPA: 3.84/4.0*
- **University of Illinois at Urbana-Champaign** — Aug. 2017 - May 2021
  Bachelor of Science in Electrical Engineering (dual degree) — *GPA: 3.87/4.0*
  Graduate with High Honors
- **Zhejiang University** — Aug. 2017 - May 2021
  Bachelor of Engineering in Electrical Engineering and Automation (dual degree) — *GPA: 3.95/4.0*
  Outstanding Graduates, Academic Excellence Award

Technical Skills

- **ML / DL:** PyTorch, VLAs ($\pi_{0.5}$, StarVLA), VLMs (Gemma, QwenVL, DreamVL), Video Models (WAN, Cosmos-predict), GNNs, Imitation Learning, Reinforcement Learning (PPO, GRPO)
- **Robotics:** ROS, real-robot deployment (UR5e, Kinova, Franka, Stretch, TurtleBot 2i), data collection & deployment of imitation learning pipeline (FastUMI, SpaceMouse, UMI gloves), simulation (Isaac Sim, MuJoCo, Sapien, Genesis)

Selected Awards

- **Best Paper / Best Student Paper Award Finalist**, **CoRL 2023**
- **Best Conference Paper Finalist**, **ICRA 2025**
- **Best Paper in Human-Robot Interaction**, **ICRA 2025**
- **Best Paper Award**, CVPR 2026 Sense of Space Workshop
- **Best Paper Award**, ICRA 2025 Workshop on Advances in Social Navigation
- **Best Poster Award**, IROS 2023 Last-Mile Robotics Workshop
- **Second Prize**, RoboMaster International Region Competition (ZJU-UIUC Team), 2019

Work Experience

- **Waymo LLC** — June 2026 - Aug. 2026
  Research Intern, Mountain View, CA
  - Researching **diffusion VLMs** for efficient long-tail autonomous-driving problems based on DiffusionGemma.
- **Stealth AI Startup** — Jan. 2026 - May 2026
  Research Intern, Remote
  - Post-trained a visual-tactile **world model** on Cosmos-Predict 2.5, validated in simulation (UniVTAC) and on an internal humanoid with a dexterous hand; reached **90%** on real-world peg insertion using a large-scale dataset collected via Sunday Robotics-style gloves.
- **Amazon Robotics** — May 2025 - Dec. 2025
  Applied Scientist II Intern, Seattle, WA
  - Delivered a transformer-based progress-prediction model (on DINOv2 features) for Vulcan Stow—the first to beat the expert heuristic (**+24% F1**); deployed to production across ∼10 robots nationwide, driving abort decisions, affordance planning, and throughput (UPH).
  - Post-trained a reasoning-augmented VLM (SFT on Qwen2.5-VL) for scene-to-affordance prediction on long-tail cases, building a chain-of-thought reasoning dataset from scratch; **+12%** affordance accuracy over the production baseline.
- **XPENG Inc.** — May 2024 - May 2025
  ML Research Intern, San Diego, CA
  - Led research on building a driving **world model** (VQ-VAE tokenizer + autoregressive Transformer, GAIA-1-style) on 2M ∼30s video clips; used for downstream planning, **+15% Driving Score** over the SOTA baseline on Bench2Drive.
  - Trained a cross-unified embodiment VLA based on Qwen2.5-VL jointly on Open X-Embodiment and internal autonomous-driving data, beating the UniAct baseline by **+10%** on LIBERO and reaching SOTA results (**82.5**) on NavSim.

Selected Publications

* Equal contribution.

- **K. Hong**, H. Chen*, J. Xu*, R. Wang*, K. Wang*, M. Zhang, S. Liu, Y. Zhu, Y. Li, and K. Driggs-Campbell, "Sim-and-Real Co-Training of Graph Neural Dynamics for Long-Horizon Scooping," in **ICRA 2026**. [website]
- **K. Hong**, S. Peng*, X. Zeng*, P. Wang*, B. Tian, H. Chen, Y. Zhu, Y. Wang, and K. Driggs-Campbell, "Self-Labeled Preference Transfer: Bootstrapping Steerable Robot Manipulation on Unlabeled Tasks," in submission.
- **K. Hong**, P. Wang, H. Jiang, X. Zeng, S. Peng, B. Tian, H. Chen, Y. Zhu, L. Sun, and K. Driggs-Campbell, "What Should a Robot's Memory Encode? Future-Target Supervision for Long-Horizon Manipulation," in preparation.
- P. Wang, **K. Hong**, C. Peng, K. Driggs-Campbell, M. Tomizuka, C. Xu, and C. Tang, "DiscreteRTC: Discrete Diffusion Policies are Natural Asynchronous Executors," in submission. [website]
- H. Chen, **K. Hong***, Y. Niu*, S. Liu, Y. Wang, Y. Li, and K. Driggs-Campbell, "Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks," in **CoRL 2023** (**Oral, Best Paper / Best Student Paper Award Finalist**). [website]
- S. Luo*, Q. Peng*, J. Lv*, **K. Hong**, K. Driggs-Campbell, C. Lu, and Y. Li, "Human-Agent Joint Learning for Efficient Robot Manipulation Skill Acquisition," in **ICRA 2025** (**Best Paper in Human-Robot Interaction and Best Conference Paper Finalist**). [website]
- S. Liu, P. Chang, Z. Huang, N. Chakraborty, **K. Hong**, W. Liang, J. Geng, and K. Driggs-Campbell, "Intention-Aware Robot Crowd Navigation with Attention-Based Interaction Graph," in **ICRA 2023** (**Best Poster Award at IROS 2023 Workshop**). [website]
- S. Liu, H. Xia, F. Pouria, **K. Hong**, N. Chakraborty, Z. Hu, J. Biswas, and K. Driggs-Campbell, "HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments," in IEEE Transactions on Automation Science and Engineering (**T-ASE**) (**Best Paper Award at ICRA 2025 Workshop**). [website]
- H. Chen, J. Xu*, H. Chen*, **K. Hong**, B. Huang, C. Liu, J. Mao, Y. Li, Y. Du, and K. Driggs-Campbell, "Multi-Modal Manipulation via Multi-Modal Policy Consensus," in **ICRA 2026** (**Best Paper Award at CVPR 2026 Workshop**). [website]

Research Projects

- **Steerable Manipulation via Self-Labeled Preference Transfer** — Jan. 2026 - May 2026
  - Proposed **Self-Labeled Preference Transfer (SPT)**, a two-stage VLA co-training a preference-conditioned action head and a preference-recognition head, then self-labeling unlabeled target demonstrations to transfer preference-following to new tasks; 96.4% recognition accuracy and **+25.4%** preference-following over a naive VLA baseline (based on StarVLA) in both simulation and challenging real-world tasks (data collected via FastUMI, inference using a wrist fisheye camera).
- **Memory-aware Long-horizon Manipulation** — Jan. 2026 - present
  - Identified a memory–action gap in long-horizon policies and proposed **future-target supervision**, a drop-in auxiliary objective training spatial memory to predict dense heatmaps of upcoming subgoals; **+25%** task success across five long-horizon RoboMME tasks over the SOTA baseline.
- **Real-Time Asynchronous Execution for Dynamic Manipulation** — Jan. 2026 - May 2026
  - Co-developed **DiscreteRTC**, showing discrete diffusion policies are natural asynchronous executors for dynamic manipulation, with **50% higher success** on real-world dynamic pick over the flow-matching Real-Time Chunking (RTC) baseline.
- **Sim-and-Real Co-Training for Graph-based World Model: Scooping** — May 2023 - Sept. 2025
  - Proposed a **sim-and-real co-training** framework with a graph-based **world model** and Monte-Carlo tree search for long-horizon granular scooping; perception from multiview calibrated RGB-D point clouds. Published at **ICRA 2026**.
- **Predicting Object Dynamics with Primitives: Stowing** — Oct. 2022 - May 2023
  - Combined GNN-based interaction prediction with primitive-augmented trajectory optimization for stowing from a single demonstration (3–4 keyframes) at **95%** real-world success on multiview RGB-D point clouds. **Best Paper Finalist, CoRL 2023**.
- **Crowd-Aware Social Navigation** — Jan. 2022 - Sept. 2025
  - Built an attention-based recurrent GNN fusing heterogeneous spatio-temporal interactions with RL and trajectory prediction, deployed on TurtleBot 2i (**ICRA 2023**); extended to **HEIGHT** for constrained environments with strong zero-shot generalization (**T-ASE**).